

Pixel Art Is a Palette, Not a Downscale: Rebuilding Targets and Decoding for Tiny Sprite Generation

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Abstract

Text-to-image models fail at the smallest images people draw on purpose: 12–24 px RGBA game sprites, where a handful of colours and hard alpha edges are the medium. We show that the usual failure is not a modelling failure but a target failure. In the sprite corpus this area trains on, only 7.1% of the images are natively 16 px or smaller; the rest is 25–64 px artwork box-downsampled into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns soft, fifty-colour blobs, and the evaluation reference, drawn from the same pool, rewards exactly that. We rebuild the targets and evaluate against references that contain only real low-resolution pixel art. Under this protocol the data work alone takes FD-DINOv2 from 167.2 to 117.9 at 16 px. We then close the remaining gap where it actually lies: real sprites use a median of six opaque colours while every sampling rule we tested emits more than fifty. Projecting the predicted $\hat{x}_0$ onto a per-sprite k -colour palette during the late denoising steps reaches 31.8 ± 0.5 — better than quantising finished samples (41.6), than one corpus-wide palette (120.1) and than a median filter that removes the same dither (112.2). The rule is a property of the recipe rather than of the network: a transformer backbone that trails the UNet by 36.5 FD without it lands within 0.7 with it. Applied to sprites from generators we did not train, it improves native FD by 36–41% while preserving 80% of the silhouette. Against external systems given the same palette post-process we reach 50.8 versus 134.4 for gpt-image-2 with downscaling.

1. Introduction

Game sprites are among the smallest images anyone draws on purpose: an item or a character on a 12–24 px RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background. At these sizes there is no texture in which to hide a mistake — one wrong pixel changes



Figure 1. Four rows of 16 px RGBA sprites, nearest-neighbour magnified, with the median number of opaque colours per row. Real sprites drawn at this size use six; the corpus’s own 16 px targets, which are 25–64 px artwork box-downsampled, use 49; gpt-image-2 rendered large and reduced uses 111; ours uses four. The third row is the easiest on the eye and the furthest from the medium.

the silhouette, one blended colour reads as blur. The obvious recipe, prompting a large text-to-image model and shrinking its output, produces something recognisable but not a sprite: it keeps the soft shading and the hundred-odd colours of an illustration (Figure 1).

This paper is about a failure one level below the model. We train a dedicated generator for this medium — a 72.5M pixel-space diffusion model over RGBA sprites, conditioned on a frozen CLIP text encoder and a resolution-bucket label — and find its weaknesses inherited from what it is asked to imitate. In the sprite corpus this area trains on, only 7.1% of the images are natively 16 px or smaller; two thirds are 25–64 px artwork that the pipeline box-downsamples into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns what they are: soft, fifty-colour blobs. Worse, the standard evaluation reference is drawn from the same pool, so it rewards the blur it caused, and the ranking of methods under it does not survive contact with real pixel art.

We change both the targets and the yardstick. On the data side we remove the duplicates, forbid feeding a sprite to a bucket more than 1.5× below its native size, and add

14,999 natively small sprites from permissively licensed collections, captioned with a vision–language model. On the evaluation side we define the *native protocol*: FD-DINOv2 against real sprites that are natively at most R px, never downsampled. At 16 px the data work alone moves the model from 167.2 to 117.9.

What remains is a property of the medium that continuous diffusion cannot express. Real 16 px sprites use a median of six opaque colours and 46% of their adjacent opaque pixel pairs are exactly equal; every sampling rule we tested emits more than fifty colours and reaches 8%, spending the difference on few-level dither inside regions that should be flat. We close that gap inside the sampler: during the late denoising steps we project the predicted $\hat{x}_0$ onto a per-sprite k -colour palette (batched k -means over the opaque pixels, alpha untouched) and continue, so the remaining steps repair the seams hard quantisation leaves. At 16 px this takes FD from 87.1 to 31.8 ± 0.5 , where quantising the finished samples reaches only 41.6.

Two results say the rule is not an artefact of our network. Replacing the UNet with a 69.7M transformer, everything else held fixed, gives a model that trails by 36.5 FD under plain guidance and by 0.7 once the projection is on: the decoding rule absorbs the difference between the two architectures. And the rule applies to sprites we did not generate — re-noising the output of gpt-image-2, FLUX.2 with a pixel-art LoRA, or SDXL with Pixel Art XL and denoising it under our recipe improves native FD by 36–41% at every resolution while keeping about 80% of the input silhouette.

We also report a negative result we consider as informative as the positive one. Cross-resolution self-guidance was, under the downscale-biased reference, our strongest method by 44%; under the native protocol it gives nothing. Which reference set one scores against decides which of these conclusions one publishes.

Contributions.

- A diagnosis of the targets used for tiny-sprite generation — native-resolution statistics, duplicate leakage, and the resulting bias of the standard reference — and a rebuilt corpus plus the native protocol that removes both.
- **Palette-projected sampling**, which lowers FD-DINOv2 at 16 px from 87.1 to 31.8 ± 0.5 over three seeds, beats post-hoc quantisation (41.6) and two controls that isolate what it is doing, and costs 0.8 points of caption retrieval at $k=6$, where the samples match the colour statistics of real sprites.
- Evidence that the rule belongs to the recipe rather than the network: it absorbs 98% of the gap between two architectures, and it improves the sprites of three systems we did not train.
- An external comparison in which every system receives the same palette post-process: at 16 px we reach 50.8

against 134.4 for gpt-image-2 with downscaling, 150.1 for FLUX.2 with a pixel-art LoRA and 244.2 for SDXL.

2. The medium and the protocol

What the corpus actually contains. Our starting corpus is the OpenGameArt-derived sprite collection used in this area: 37,489 unique RGBA sprites after cutting sheets into connected components and filtering. Its low-resolution buckets are not made of low-resolution art. Only 7.1% of the sprites are natively 16 px or smaller and 33% are natively 24 px or smaller; two thirds are 25–64 px artwork that the training pipeline box-downsamples into whatever bucket it is asked for. The consequences are measurable in the targets themselves: a real sprite drawn at 16 px has a median of six opaque colours and 46% of its adjacent opaque pixel pairs are exactly equal, while the corpus’s own 16 px targets have 49 colours and are soft (Figure 1, rows 1–2).

Duplication compounds it. Hashing every sprite by its exact pixel content finds 4,888 extra copies, and because hold-out sets in this area are defined by path, a pixel-identical twin of a held-out sprite stays in training: 16.2% of the 3,000 evaluation sprites have one, counting a twin anywhere training is allowed to see. We drop every extra copy and every copy of a held-out sprite.

Rebuilt targets. Three changes: remove the duplicates; forbid feeding a sprite to a bucket more than $1.5\times$ below its native size, so a 64 px illustration can no longer become a 16 px target; and add 14,999 natively small sprites cut from three permissively licensed OpenGameArt collections (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0), captioned with a vision–language model and filtered for fonts, UI elements and near-identical animation frames. Per-file licence records accompany the added material.

The native protocol. The reference set decides the result, so we define it explicitly. *Native* reference: all real sprites that are natively at most R px, never downsampled — 2,675 of them at 16 px, split in half so a real-versus-real floor can be computed on a disjoint set (4.47 at 16 px). *Biased* reference: the conventional one, a random 3,000 of the corpus, of which 92.9% is downsampled artwork at 16 px (floor 3.45). We report both throughout, because the two rank methods differently and the disagreement is part of the result.

Scores are Fréchet distances of DINOv2-small CLS features [17, 22] computed at the native resolution: sprites are composited on white and nearest-neighbour upsampled to 224 px. One sample per caption for 3,000 held-out captions, which are recaptions of the held-out sprites, so the prompt distribution matches the data. Seed-to-seed standard deviation of our headline configuration is 0.49 over three seeds, and an independent retraining of the same recipe

lands within 2.5 FD, so we do not read differences below that scale. We also report CLIP ViT-B/32 image–text similarity and retrieval of the own caption among 100, with the distractor draw re-seeded per set.

Leakage control. Exact duplicates are removed, but near-duplicates are not: by an 8×8 luminance-plus-alpha average hash, 1,241 of the 3,000 held-out prompts (41.4%) still have a twin in training. Rescoring every configuration on the 1,759 prompts that do not moves each number by at most 3.0 FD — the largest single change is 167.2 to 164.2 — and changes no ordering, so the main tables use the full set.

3. Related work

Pixel art. Most work on pixel art converts an existing image rather than generating one: joint palette-and-pixel optimisation [7], contrast-aware downscaling [26], learned pixelization [9, 25], and optimisation-based stylisation with a diffusion prior [4]. Text-to-sprite generation in practice is served by community LoRAs on large models [15, 16], which have no published description of their data or training. Palette-conditioned generation has been proposed [14], with the palette supplied as an input; we estimate it per sprite from the model’s own prediction and need no extra conditioning.

Guidance. CFG [10] extrapolates away from an unconditional prediction; a long line of work reshapes that update [6, 13, 19, 21] or replaces the reference with a weaker prediction from the same network — a smaller model [12], perturbed attention [1, 11], or a random condition [20]. We use independent-condition guidance, which needs no unconditional training, and show it is the rule that survives our change of reference set.

Constraints inside the sampler. Projecting the predicted $\hat{x}_0$ at every step is the standard way to impose a hard constraint on a diffusion model [5, 23]; discrete-state formulations impose quantisation by construction instead [2, 8]. Our projection is that device applied to the property that defines this medium, and it changes nothing about training.

Reference sets decide rankings. That a benchmark’s reference set can decide an ordering is known in generative evaluation, from FID’s sensitivity to the feature extractor and preprocessing [18, 22] to duplicate contamination in image corpora [3, 24]. Our instance is concrete: for tiny sprites the conventional reference is 92.9% downscaled artwork, and that single fact reverses the ranking of the guidance methods we would otherwise report.



Figure 2. Palette-projected sampling at 16 px, same prompts and seed. Without projection (top) flat regions carry a few levels of dither; projecting the predicted $\hat{x}_0$ onto four or six colours during the late steps (middle, bottom) removes it and the remaining steps repair the seams. Native FD 87.2 / 36.9 / 34.1.

4. Palette-projected sampling

The model. A pixel-space diffusion model over RGBA sprites: a 72.5M UNet with cross-attention on a frozen CLIP ViT-B/32 text encoder and a learned embedding of the *resolution bucket* $b_R \in \{12, 16, 20, 24, 32, 48, 64\}$, trained with the usual ϵ -objective on a squared-cosine schedule. A sprite is centred on an $R \times R$ canvas with hard alpha. Nothing here is novel; it is the smallest model that lets us study the targets and the decoding rule.

The rule. A 16 px sprite is defined as much by how few colours it uses as by its shape, and a continuous denoiser in RGB has no mechanism that snaps a region to one value. At every step with $t/T \leq f$ we form the current estimate $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \tilde{\epsilon}) / \sqrt{\bar{\alpha}_t}$, cluster its *opaque* RGB pixels into k groups with a batched k -means (8 iterations, initialised at evenly spaced luminance quantiles), replace each pixel by its centroid, leave the alpha channel and transparent pixels untouched, and map the result back to a noise prediction:

$$\hat{x}_0^{\text{pal}} = \Pi_k(\hat{x}_0), \quad \tilde{\epsilon} \leftarrow (x_t - \sqrt{\bar{\alpha}_t} \hat{x}_0^{\text{pal}}) / \sqrt{1 - \bar{\alpha}_t}, \quad (1)$$

before taking the scheduler step. The projection is a hard constraint on the model’s own belief rather than a post-process: the steps that follow see the quantised state and repair the seams it creates, which is why doing it inside the sampler beats quantising the finished image (31.8 against 41.6 at 16 px). It adds no network evaluations; k -means on at most 576 pixels is free next to a UNet pass. We use $f = 0.5$; the optimum is flat over $[0.4, 0.5]$ and running the projection for the whole schedule costs two points, so the timing matters far less than the projection.

Table 1. 16px, 3,000 held-out captions. *Native*: reference of sprites natively ≤ 16 px (floor 4.47). *Biased*: the conventional corpus-sample reference (floor 3.45), 92.9% of it downsampled artwork. The two columns rank the same methods in opposite orders.

Targets	Sampler	Native ↓	Biased
original	CFG 1.5 (best)	167.2	11.1
policy fixed	CFG 1.5	123.1	14.8
+ new sprites	CFG 1.5	117.9	13.6
+ new sprites	cross-res. ref., $w=1.5$	114.9	14.2
+ new sprites	internalised, 1 pass	170.3	10.4
+ new sprites	ICG $w=2$ [20]	87.1	20.7
+ new sprites	ICG + post-hoc 4-colour	41.6	111.0
+ new sprites	ICG + projection, $k=6$	36.9	71.1
+ new sprites	ICG + projection, $k=4$	31.8 ± 0.5	111.2

5. Experiments

Main result. Table 1 separates the three things that changed. Rebuilding the targets moves the model from 167.2 to 117.9 under the native protocol without touching the sampler. Independent-condition guidance takes it to 87.1, and palette projection to 31.8 ± 0.5 over three seeds — a larger step than the entire distance between guidance rules. Projecting inside the sampler is what makes that step work: quantising the finished samples to the same four colours reaches only 41.6, because nothing repairs the seams quantisation introduces. The biased column moves in the opposite direction throughout, which is the disagreement Section 7 is about.

Is it the palette, or just fewer colours? Two controls replace Π_k at exactly the same steps. One projects onto a *single* four-colour palette shared by every sprite, learned by k -means over the reference set: the same colour count without the per-sprite structure. The other applies a 3×3 median filter, which removes the same dither and imposes no palette. Both are *worse* than not projecting at all — 120.1 and 112.2 against 87.1 — so the gain is neither “fewer colours” nor “smoother”, but each sprite’s own small palette.

Palette size, with a validation split. Sweeping k at 16px gives 38.1 / 31.8 / 33.7 / 36.9 / 50.6 for $k = 3/4/5/6/8$, and 39.9 when k is drawn per sprite from the corpus histogram. Because that sweep and the reported number share a split, we also split the captions in half by index: choosing on one half selects the same $k=4$, which scores 33.36 on the disjoint half against 38.63 / 35.62 / 38.27 / 43.76 for the alternatives (values at $n=1500$ are not comparable to $n=3000$ ones, as FD is biased by sample size). The FD optimum $k=4$ is nonetheless *below* the real median palette: measured on the samples, $k=6$ is the setting that reproduces the medium

— six opaque colours and 43% exactly-equal neighbours, against six and 46% for real sprites — while $k=4$ gives four colours and 54%, flatter than the real thing. We report $k=4$ as the best-FD setting and $k=6$ as the one to use.

Prompt alignment. Projection trades a little text faithfulness for a large distributional gain. CLIP 100 cos / retrieval@1 among 100 are 27.60 / 22.8% for ICG alone, 27.38 / 22.0% at $k=6$ and 27.07 / 19.2% at $k=4$, against 27.66 / 22.6% for the real held-out sprites the captions were written from: at $k=6$ our samples are as retrievable as real sprites. The cost is smaller on a larger canvas but stops shrinking — 3.6 points at 16px, 1.9 at both 20 and 24px.

Is it the network? Every number above comes from one convolutional network. We retrained the 20k-step pilot with the UNet replaced by a 69.7M-parameter transformer (adaptive-norm DiT; the bucket enters through the text context, as that design has no class-conditioning path), leaving corpus, schedule, sampler and evaluation untouched. At 20k steps the transformer is the weaker model by a wide margin — 185.6 against 149.1 native FD under plain CFG — and yet under the reported configuration the two land at 42.4 and 43.1 (two seeds each, spread 0.3 and 0.4). The projection absorbs 98% of the difference between the two architectures, so it is a property of the recipe and not of convolutions.

More data, and why a short pilot cannot tell you whether it helps. The corpus is only 7.1% natively small, so more natively small sprites is the obvious remedy. We found two sources: the Kenney CC0 packs (6,566 usable rows, a median palette of three colours, but only 41.8% whole objects) and the OpenGameArt CC-BY-3.0 bundle our rebuild had never touched (11,988 usable rows against which exact pixel hashing finds 13 duplicates, 82.4% whole objects), together taking training from 86,290 rows to 118,378. At 20k steps the addition looks decisive: the reported configuration moves from 54.2 to 53.2 with the Kenney packs and to 42.9 with the CC-BY-3.0 bundle, a 19.4% gain and twenty times the seed spread. Trained to the same 80k steps as every other model here, the enlarged corpus is *worse*: 39.3 ± 0.3 against 31.8 ± 0.4 over three seeds each, with plain CFG unchanged. The ordering at 20k is the reverse of the ordering at 80k, and only the second is the model anyone would ship. We offer no mechanism, because the obvious one does not survive measurement — the natively-small fraction of the added material is 24.0% against 28.6% for what was already there, nowhere near enough to explain a 24% regression.

This is the second time in this work that a 20k-step comparison inverted by 80k; the first was capacity, where a $2.2 \times$ larger network led at 20k (46.1 against 54.2), had lost by 40k (38.3 against 35.8) and was clearly behind at 80k (38.5 against 31.8). Short pilots are how this kind of study is made

Table 2. External comparison, 200 held-out captions, native protocol. Every row is scored after the same k -colour post-process ($k = 4/8/6$ at 16/20/24 px); ours applies it inside the sampler instead. “Real” are held-out sprites of the original, mostly down-scaled corpus.

System	16 px	20 px	24 px
Ours	50.8	152.5	192.2
Real sprites (mixed provenance)	77.0	165.5	150.7
gpt-image-2 + downscale	134.4	448.6	478.0
FLUX.2-klein + pixel-art LoRA	150.1	473.9	472.4
SDXL + Pixel Art XL	244.2	526.3	527.2

affordable, and in this setting they are safe for rejecting a change and not for accepting one. Every headline number here is measured at full length for that reason.

Other resolutions. The recipe transfers and the useful palette size grows with the canvas. Native FD with ICG $w=2$, three seeds at the best k of each resolution: 143.2 ± 0.8 at 12 px ($k=8$), 31.8 ± 0.5 at 16 px ($k=4$), 96.7 at 20 px ($k=8$) and 124.2 at 24 px ($k=6$). The absolute values are not comparable across resolutions — each has its own reference set and floor.

External systems. Table 2 compares three publicly available pipelines on 200 held-out captions, each generating at its native size and reduced to the sprite canvas by the same code we use for real data, with the same palette post-process. Without that post-process they score $284.4 / 318.9 / 436.4$ at 16 px, so the palette step helps them too; they remain $2.6\text{--}4.8\times$ further from real pixel art than our model at every resolution. At 16 and 20 px we are also closer to native pixel art than the mixed-provenance sprites the conventional protocol calls ground truth, which is a statement about that corpus rather than about our model. At 24 px the real sprites are still ahead of us.

6. The rule on other systems

Quantising a finished external sample helps but cannot repair the seams quantisation leaves, the same limit we measured on our own samples. The projection can nevertheless be applied to another system’s output: re-noise its sprite to a fraction σ of the schedule and denoise it under our configuration. Table 3 reports both halves — how much closer to real pixel art the result gets, and how much of the original survives — because at $\sigma=1$ the procedure is simply our own model, and a distributional score alone cannot tell repair from replacement. Retention is measured in pixel space, as the DINOv2 cosine is saturated at this size: two unrelated sprites for the same caption already score 0.918.



Figure 3. Eight captions drawn at random. Rows: gpt-image-2 rendered large and reduced with the same palette post-process as every other row; the same sprite re-noised to σ of the schedule and denoised by our model; our model generating the caption from noise; and the held-out sprite. The refined sprites keep the subject and also look better than our own samples for the same captions, which score 86.5 and 50.8 native FD respectively: what our model still lacks is not the decoding rule but the ability to draw the subject.

Table 3. The decoding rule applied to external sprites, 200 captions, 16 px. σ is the fraction of the schedule re-run. Retention is measured against the system’s own output; the floor is our model generating the same captions from noise, i.e. what “no retention” looks like.

System	Native FD ↓			Retention
	post-hoc	$\sigma=0.4$	$\sigma=0.6$	IoU @ 0.6
gpt-image-2	134.4	115.9	86.5	0.788
FLUX.2 + LoRA	150.1	116.1	88.2	0.803
SDXL + Pixel Art XL	244.2	197.2	155.0	0.813
Ours, from noise		50.8		0.513

At $\sigma=0.6$ about 80% of the silhouette survives, well above the 0.46–0.53 floor of a freshly generated sprite, and native FD improves by 36–41% for all three systems; the same holds at 20 and 24 px (448.6 to 265.7 and 478.0 to 299.0). Sweeping σ on gpt-image-2 gives a monotone trade-off with no knee, and the endpoint is a self-check: at $\sigma=1$ the run reproduces our own samples on all three measurements — FD 53.6 against 50.8, retention 0.518 against 0.513, retrieval 16.0% against 15.0%. Caption retrieval falls from 22.0% for gpt-image-2’s own sprite to 18.5% after refinement and 15.0% when our model draws the caption itself, so refinement keeps most of a large generator’s semantic advantage while moving the style towards the medium. It does not close the gap to generating natively, and the gap widens with resolution: a post-process fixes how a sprite is drawn, and the larger the canvas the more of the remaining distance is structure it cannot touch.

7. Discussion and limitations

Why the two protocols disagree. The biased reference is 92.9% downscaled artwork, so it is a reference for *blurred illustrations*, and the sampler that scores best against it is the one that produces them. That is why heavy guidance wins there and loses here, why cross-resolution self-guidance — which makes samples look like a lower-resolution version of themselves, i.e. like downscaled art — wins by 44% there and gives nothing here, and why palette projection, which moves samples from 52 colours to four, improves the native score by a factor of 2.7 while making the biased score five times worse. Neither number is wrong; they measure distance to different populations, and only one of those populations is pixel art.

Can a model be asked instead of people? We put the same pairs to a vision–language model: 560 forced choices over 40 captions, with the real held-out sprites in the pool as a control. The control fails. Asked which sprite looks more like real low-resolution pixel art, the judge puts FLUX.2 and gpt-image-2 above the real sprites themselves (72% and 70% head to head); replacing the question with the concrete distinction — drawn pixel by pixel at this size, or a larger picture shrunk down — moves the real sprites from 40.6% to 48.8% and the blurriest baseline from 41.2% to 27.5%, but FLUX.2 still beats the real sprites 68% of the time. Our own samples tie with real sprites at 50% under both wordings. We report this as a negative result about the instrument: whether an image was authored at sprite resolution is not something a general-purpose vision model recovers from the image, which is the work the native reference set is doing.

Limitations. *Scope.* Everything is measured on models we trained, on one family of sprite corpora; the transformer run rules out convolutions as the explanation but not a shared bias of pixel-space diffusion, and we have not tested a second domain. *Metric.* FD with DINOv2 on 12–24 px images upsampled to 224 px is sensitive to exactly the colour and contrast statistics the projection changes. Inception clean-FID on the same samples and reference preserves the ordering, including both controls, but compresses it — a 2.8× gap becomes 1.10× — because an encoder that resizes to 299 px cannot see hard edges at 16 px. We have no human study, and it remains the obvious next check. *Tuning.* k transfers across a validation split; for f the honest statement is that the optimum is flat over $[0.4, 0.5]$ rather than validated. *Where we still lose.* At 24 px real sprites remain closer to native pixel art than our samples, and a large generator plus our post-process still draws subjects better than we generate them.

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